

Math 526, F06
Quiz 10

Name:
SSN: xxx-xx-

Instructions: There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & -2 & 1 \end{bmatrix}$. With the starting vector $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, compute the first four loops of the power method to find an approximation of the dominant eigenvalue of the matrix $\mathbf{A}$.

Solution:

1st loop:

$$\mathbf{z} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} = \begin{bmatrix} 0.7071 \\ 0 \\ 0.7071 \end{bmatrix}, \quad \mathbf{x} = \mathbf{A}\mathbf{z} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 0.7071 \\ 0 \\ 0.7071 \end{bmatrix} = \begin{bmatrix} 2.1213 \\ -1.4142 \\ 0.7071 \end{bmatrix},$$

$$\alpha = \mathbf{z}^t \mathbf{x} = 2.$$

2nd loop:

$$\mathbf{z} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} = \begin{bmatrix} 0.8018 \\ -0.5345 \\ 0.2673 \end{bmatrix}, \quad \mathbf{x} = \mathbf{A}\mathbf{z} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 0.8018 \\ -0.5345 \\ 0.2673 \end{bmatrix} = \begin{bmatrix} 1.8708 \\ -1.0690 \\ 1.3363 \end{bmatrix},$$

$$\alpha = \mathbf{z}^t \mathbf{x} = 2.4286.$$

3rd loop:

$$\mathbf{z} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} = \begin{bmatrix} 0.7379 \\ -0.4216 \\ 0.5270 \end{bmatrix}, \quad \mathbf{x} = \mathbf{A}\mathbf{z} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 0.7379 \\ -0.4216 \\ 0.5270 \end{bmatrix} = \begin{bmatrix} 2.2136 \\ -1.4757 \\ 1.3703 \end{bmatrix},$$

$$\alpha = \mathbf{z}^t \mathbf{x} = 2.9778$$

4th loop:

$$\mathbf{z} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} = \begin{bmatrix} 0.7397 \\ -0.4931 \\ 0.4579 \end{bmatrix}, \quad \mathbf{x} = \mathbf{A}\mathbf{z} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 0.7397 \\ -0.4931 \\ 0.4579 \end{bmatrix} = \begin{bmatrix} 2.1486 \\ -1.4089 \\ 1.4441 \end{bmatrix},$$

$$\alpha = \mathbf{z}^t \mathbf{x} = 2.9454$$